

UNIT 11: INITIAL RESPONSE AND SEARCH PLANNING

I. INTRODUCTION

The identification of the ship in distress plays an important role in organizing the implementation of the search and rescue plan and directly affects the results of the search process for the ship in distress. The following lesson will introduce the search and rescue planning process in accordance with international regulations.

The Vietnam Coast Guard ship 8005 rescue ship in distress



II. TEXT

A. Initial Response and Search Planning

Effective, efficient prosecution of a the search and rescue (SAR) incident requires well thought out procedures. Not every incident will develop into a full SAR case, but every case has the potential to greatly expand. This guidance provides

processes necessary for a rapid and thorough reaction upon receiving notification of a potential or actual distress.

B. Initial Response

Upon receiving a SAR incident, the initial response datum shall be quickly established. In the interest of saving time and effort when doing drift computations manually, the datum for the initial response may be determined by calculating drift using the object's last known position and the effects of water current and wind without considering leeway divergence (Figure 11-1). Time of datum must take the underway and transit times for the search and rescue unit (SRU) into consideration.

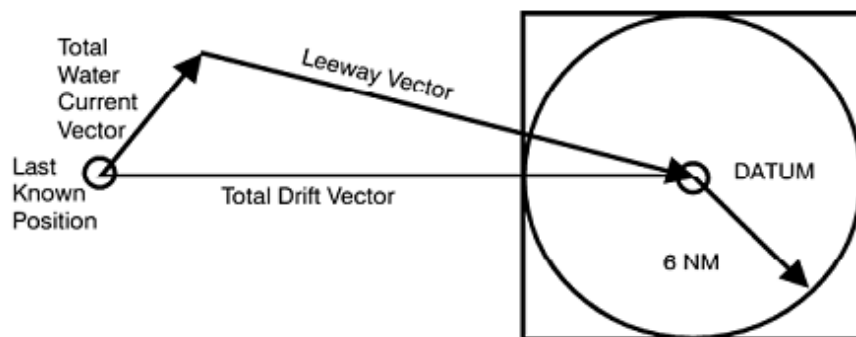


Photo: Vessel Adrift (Quick Calculation for Initial Response)

The purpose of drawing a radius around the datum is to describe the geographical area most likely to contain the search object. The formula uses the Total Probable Error (E), which includes the Initial Position Error (X) of the distressed craft and the Navigation Error of the SRU (Y), but does not include Drift Error (De). For coastal SAR cases when an SRU arrives on scene and the search object is not seen, a 5 nm position error for the distressed craft and a 1 nm navigation error for the SRU may be assumed. Using the formula, when X = 5 and Y = 1 then:

$$E = \sqrt{X^2 + Y^2}$$

$$E = 5.009 \text{ NM}$$

Applying the safety factor for the first search $f_s = 1.1$, then

$$R = E \times fs \text{ or } R = 5.6NM$$

This is round up to 6 NM

If the initial response SRU reports arriving on scene without finding the search object, the Search And Rescue Mission Co-Ordinator (SMC) shall develop a more comprehensive search plan and shall notify appropriate additional resources that they may be needed and may deploy some of them immediately if conditions warrant. Examples of issues to consider include, but are not limited to, the survival prospects of the distressed person(s), remaining daylight hours, remaining endurance of the initial response SRU, etc.

C. Search Planning

Search planning consists of determining where to deploy the available search facilities, developing an attainable search plan, selecting search patterns, planning on scene coordination, developing a Search Action Plan (SAP) containing specific SRU assignments and all necessary coordination details, transmitting the complete SAP to the on-scene coordinator (OSC) and all participating search facilities, and reviewing the search results. Many factors influence the movements of the search object and the range of its possible locations as of the time when search facilities can be on scene. The SMC judges the impact of these factors to determine the region to search and methods to use, evaluates the number and capabilities of available search facilities, and tries to increase the probability of success (POS) as quickly as practicable with those search facilities.

Search planning involves the following steps:

- (a) Evaluate the situation, including any previous search results;
- (b) Estimate the possible distress incident positions
- (c) Estimate the survivors' post-distress movements to produce an estimate of their possible locations and associated probability densities at the time when search facilities can be on scene;
- (d) Determine the best way to deploy the available search facilities so the chances of finding the survivors are maximized;

- (e) Define search sub-areas and search patterns for assignment to specific search facilities;
- (f) Provide a comprehensive search action plan to the OSC, all search facilities, and other appropriate parties that includes a current summary of the situation, detailed search object description(s), specific search assignments for the search facilities, on scene coordination instructions, and search facility reporting requirements.

Since objects in the marine environment tend to drift under the influence of wind and current, it is necessary to estimate the search object's location when planning a search, which is done by updating the datum position(s) to account for drift. The size of the search area depends on the size of the total probable error of position (E) and on the amount of searching effort that is available. The size of the total probable error of position depends on the probable error of the initial distress position (X), the probable error due to uncertainty in the drift estimate (De), and the probable error of the search craft's position while searching (Y). The uncertainty of the drift estimate depends, in turn, on the uncertainties associated with the environmental factors affecting drift (wind and current) and the limitations in our knowledge of how drifting objects respond to these environmental factors.

The total probable error of position (E) is calculated from the formula:

$$A = \sqrt{X^2 + Y^2 + D_e^2}$$

Initial Position Error (X) of the distressed craft and Search Craft Position Error (Y) are the estimated errors of position based on navigational accuracy of the distressed craft and of the search assets.

If the information on the means of navigation to be used by the distressed craft or by a search facility is available, the navigational fix errors listed in table may be used for positions reported as navigational fixes.

III. GRAMMAR

1. Past Perfect

1.1. Use

- To express an action in the past that happened before another action the past.

E.g. When I got home, John had cooked a meal.

They hadn't taught English before they came here.

* Note:

Notice the use of the Past Perfect and the Past Simple in the following sentences:

+ When I got home, John cooked a meal.

+ When I got home, John had cooked a meal.

1.2. Form

a) Positive

S+ had + P_{II}

b) Negative

S+ hadn't + P_{II}

c) Questions

Had + S + V_{II}?

Yes, S had / No, S hadn't

d) Wh-question

Wh+had+S+ P_{II}?

2. Reported statements

Direct speech	Reported speech
Present Simple + "I love you". Present Continuous + "I am going out now". Present Perfect + "We have met before". Past Simple + "We met in 1987". Will + "I'll mend it for you". Can + "I can swim".. May + "We may not come". Have to + I have to go now. Must + My friend must wear glasses.	Past Simple + He said he loved me. Past Continuous + Ann said she was going out then. Past Perfect + She said they had met before. Past Perfect + He said they had met in 1987. Would She said that she would mend it for me. Could + She said she could swim. Might + They said that they might not come. Had to + She said that she had to wear glasses. Had to + She said that her friend had to wear glasses.

IV. VOCABULARY

1	SAR facility	/ɛs-eɪ-ɑː fə'sɪləti/	n	Tìm kiếm và cứu nạn
2	Track spacing	/træk 'speɪsɪŋ /	n	Theo dõi khoảng cách
3	Sweep width	/swi:p wɪðθ/	n	Chiều rộng quét

4	Datum point	/ˈdeɪtəm pɔɪnt/	n	Điểm bắt đầu trôi
5	Dead reckoning	/dɛd ˈrɛkənɪŋ/	n	Điểm chết
6	Drift	/drɪft./	n	Sự trôi dạt
7	Initial position error	/ɪˈnɪʃəl pəˈzɪʃən ˈerə/	n	Sai số vị trí ban đầu
8	Last known position	/lɑːst nəʊn pəˈzɪʃən/	n	Vị trí biết lần cuối
9	Leeway	/ˈliːweɪ/	n	Sự trôi dạt
10	Leeway divergence	/ˈliːweɪ daɪˈvɜːdʒəns/		
11	On-scene endurance	/ɒn-siːn ɪnˈdjʊərəns/		
12	Search pattern	/sɜːʃ ˈpætən/		
13	computations	/ˌkɒmpjəˈteɪʃnz/	n	Sự ước tính
14	initial	/ɪˈnɪʃəl/	Adj	Tính ban đầu
15	eographical			
16	attainable	/əˈteɪnəbəl/	Adj	Có thể đạt tới
17	Estimate	/ˈɛstɪmɪt/	N (v)	Sự đánh giá Ước lượng

18	Determine	/dɪ'tɜːmɪn/	v	Xác định, quyết định
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V. PRACTICE

Exercise 1. Read the text again and answer the questions

- Describe steps to plot Total Water Current on the chart
- Describe how to plot the Datum left, Datum Right and Search Area?
- How to establish initial response datum quickly?

Exercise 2. Rewrite these following sentences

- “The mission will be a success,” said the general.
→ The general.....
- “The squad is preparing for inspection,” reported the sergeant.
→ The sergeant reported
- “The ship will be at sea tomorrow,” announced the captain.
→ The captain announced
- “The fleet is currently in formation,” stated the admiral.
→ The admiral
- “The regiment will be on the march next week,” told the colonel.
→ The colonel
- “The platoon is currently on patrol,” informed the lieutenant.
→ The lieutenant
- “The battalion will be in the field soon,” declared the major.
→ The major

8. "The tank will fire at the target," said the soldier.

→ The soldier

9. "The helicopter is landing on the aircraft carrier," reported the pilot.

→ The pilot

10. "The jet will fly at an incredible speed tomorrow," announced the airman.

→ The airman announced

Exercise 3. Fill the gap with correct verb

1. The general said that the mission (be)_____ a failure.
2. The sergeant reported that the squad (be)_____ ready for inspection.
3. The captain announced that the ship was at sea.
4. The soldier said that the tank (fire)_____ at the target.
5. The pilot reported that the helicopter (land)_____ on the aircraft carrier.
6. The airman announced that the jet (fly)_____ at an incredible speed.
7. The sailor stated that the submarine (dive) _____ to a great depth.
8. The officer told that the missile was (launch) _____ from an underground silo.
9. The marksman informed that the rifle (fire)_____ at the target.
10. The trooper declared that the grenade (explode)_____ with a loud bang.

Exercise 4. Chose the correct answer

1. The pilot told that the helicopter _____ on the aircraft carrier.
a) had landed b) landed c) landing
2. The seaman informed that the crew _____ the deck.
a) swabbed b) has swabbed c) had swabbed

3. The commander declared that the mission _____ a success.
a) are b) had been c) was
4. The ensign said that the anchor _____.
a) had been dropped b) was dropped c) dropped
5. The lieutenant reported that the supplies _____ on board.
a) were loading b) had been loaded c) were loaded
6. The navigator announced that the course _____ plotted.
a) had been b) has been c) was
7. The professor said that the lecture _____.
a) have been cancelled b) cancelled c) had been cancelled
8. The student reported that the assignment _____.
a) submitted b) had been submitted c) was submitted
9. The dean announced that the graduation ceremony _____.
a) had been postponed b) are postpone c) was postponed
10. The advisor stated that the course schedule _____.
a) was changed b) been changed c) had been changed
11. The librarian told that the book _____.
a) returned b) had been returned c) returne
12. The researcher informed that the experiment _____.
a) completed b) had been completed c) was completed
13. The administrator declared that the application _____.
a) had been processed b) processed c) processing
14. The instructor said that the exam _____.
a) is graded b) has been graded c) had been graded
15. The graduate reported that the thesis _____.

a) had been defended b) has been defended c) was defended

16. The registrar announced that the transcripts _____.

a) has been sent b) had been sent c) were sending

Exercise 5. Work in pairs. Would you like to play this game?

Why?/Why not? Use the adjectives, verbs and phrases below to help you.

Adjectives

addictive boring difficult exciting

exhausting healthy time-consuming

Explaining preference

I'd find it...

It sounds really...

I'd like to play it because...

It appeals/doesn't appeal to me because...

I'm (not) really into...

I don't mind but...

I'd rather...

I can't stand...

Share your opinions with the class.